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APPLIED RESEARCH

Advancing Tinnitus Therapeutics: GPT-2 Driven Clustering Analysis of Cognitive Behavioral Therapy Sessions and Google T5-Based Predictive Modeling for THI Score Assessment

YONGWOO JEONG¹, (Member, IEEE), JAE-JUN SONG^{2,3}, JISEON YANG¹, AND SUNGMIN KANG¹

¹Rowan Inc., Seoul 03145, Republic of Korea

²Neurive Company Ltd., Gimhae-si 50969, Republic of Korea

³Department of Otorhinolaryngology-Head and Neck Surgery, Korea University Medical Center, Seoul 02841, Republic of Korea

Corresponding authors: Yongwoo Jeong (oscar.eaglewatch@gmail.com) and Jae-Jun Song (jjsong23@gmail.com)

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ABSTRACT Cognitive Behavioral Therapy (CBT) for tinnitus alleviates psychological discomfort caused by severe tinnitus symptoms. During CBT, the patients will have various homework assignments, including writing daily diaries and self-monitoring. Most of these homework assignments are hand-written, textual data. This paper proposes that tinnitus therapeutics can utilize Large Language Models (LLMs) to analyze CBT and predict the outcomes of CBT treatments to manage high caseloads. We anonymized patient data and examined it with GPT-2-based-embedding, dimensionality reduction, and clustering process to observe how patients themselves changed their misconceptions and developed less unnecessary excessive emotional discomfort and how their Tinnitus Handicap Inventory (THI) scores were improved after the CBT treatment. We also discussed clustering results as a part of the demonstrations that LLMs can give us insights into the CBT. Then, we augmented textual patient data in three ways to minimize augmentation bias with a corresponding penalty to overcome the constraints of limitation of the number of datasets. We trained the Google T5 Transformer with the augmented data to predict the THI score outcomes at the end of the CBT sessions. We measured the performance using the ROUGE-L metric during the training and validation. The generated THI scores by Google T5 were converted from strings to floats to measure RMSE performance, which proved that the LLM could predict the outcome of CBT treatment with CBT data. Even though there is a risk of overfitting issues, this work demonstrated that tinnitus therapeutics experts can employ LLMs to manage caseloads.

INDEX TERMS Augmentation, cognitive, CBT, GPT-2, Google, tinnitus, T5, RMSE, ROUGE-L.

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I. INTRODUCTION

Tinnitus is a prevalent medical symptom, and contributing risk factors encompass hearing loss, the use of ototoxic medications, head injuries, and depression [1]. The population of tinnitus patients has been growing. Multiple factors can

cause tinnitus. The most known reasons why most people have tinnitus are the following: natural cause (aging), ear-wax blockage, ear and sinus infections, noise-induced hearing loss, medication, medical conditions, head or neck injuries, excessive alcohol or caffeine intake, stress, anxiety, smoking habits, and long-time exposure to loud noises (earphones). Since the sources of tinnitus are various, and there is no universal treatment, the treatment must be adaptive. The physicians may choose sound therapy, cognitive behavioral therapy, tinnitus retraining therapy, medication, biofeedback, acupuncture, hypnotherapy, or more, depending on the patient's symptoms. Regarding cognitive behavioral therapy (CBT) for tinnitus, tinnitus was not the first medical condition physicians considered to use CBT for it. Aaron Temkin Beck, a professor in the Department of Psychiatry at the University of Pennsylvania, is known to be a founder of cognitive therapy and cognitive behavioral therapy. Even though he published a book in 1976 that mainly covered depression patients, the basic concept of cognitive therapy was well explained; he described discussions of how the hypothesis made by depression patients became wrongfully valid by themselves and how negative concepts of the patients can be adjusted and corrected by the cognitive approach for counteracting depression consists of using strategies that enable the patients to respect themselves positively rather than disrespect themselves negatively [2]. Since the symptom of depression is also a substantial life-changing factor among tinnitus patients, practicing CBT for tinnitus patients is recommended and covered by many research papers [3], [4], [5].

Many researchers are exploring using artificial intelligence (AI) in biomedicine to develop treatments for tinnitus. They mainly analyze electroencephalographic (EEG) signal data and use AI algorithms, such as Convolutional Neural Networks (CNN), Multilayer Perceptron (MLP), fully connected neural networks, Support Vector Machine (SVM), Naive Bayes (NB), Linear Discriminant Analysis (LDA), and Artificial Neural Network (ANN), to diagnose and predict the effectiveness of tinnitus treatment. For instance, Doborjeh et al. used CNN and MLP algorithms to predict the effectiveness of tinnitus treatment using EEG signal data [6]. Similarly, Allgaier et al. focused on tinnitus diagnosis using fully connected neural networks based on EEG signal data [7]. Manta et al. used four classification algorithms to analyze the signals of tinnitus patients through Auditory Brainstem Response (ABR) and Auditory Middle Latency Response (AMLR) to classify the data [8]. Furthermore, Rodrigo et al. predicted the results of Internet-based cognitive behavioral therapy using ANN and SVM algorithms using numerical data from patient records [9]. Lin et al. suggested a Multimodal Multi-tasking Convolutional Network (MMCN), which labels tinnitus and predicts its severity by customizing ResNet-18 deep neural networks [10]. MMCN understands multimodal neuroimaging data - T1 weighted and T2 weighted images. Their work focuses on brain MRI scan images. Furthermore, there has been an endeavor to apply natural language processing

AI techniques for tinnitus document classification. Li et al. leveraged the Bidirectional Encoder Representations Transformer (BERT) to determine tinnitus with actionable radiology reports [11]. To harness AI technology, copious amounts of data are needed in general. However, data augmentation might be the solution for scarce data availability. Also, the data augmentation avoids overfitting of the AI models and enhances robustness. Several attempts at effective text data augmentation have been conducted [12]. Wei and Zou presented a text augmentation called Easy Data Augmentation (EDA) for text classification [13]. EDA consists of four operations, which are Synonym Replacement (SR), Random Insertion (RI), Random Swap (RS), and Random Deletion (RD). Yu et al. proposed Hierarchical Data Augmentation (HDA). Key components of HDA are hierarchical attention networks that extract crucial information from text data by utilizing attention mechanisms [14]. Kumar et al. investigated Transformer-based models, GPT-2, BERT, and BART, for conditional text augmentation [15]. This study aims to explore the innovative potential of Large Language Models (LLMs) for directly addressing tinnitus, marking a significant departure from prior AI-based methodologies that predominantly focused on analyzing numerical data from EEGs, brain MRI scans, and patient records or on offering indirect CBT-based therapeutic interventions. Utilizing LLMs to analyze textual data from CBT, namely home assignments, our methodology seeks to uncover predictive patterns and themes that could more effectively inform treatment strategies, offering patients personalized and actionable feedback. Furthermore, our study proposes the hypothesis that it is also possible to predict the Tinnitus Handicap Inventory (THI) score by utilizing LLM, thereby facilitating more customized and efficacious treatment options. Applying LLMs to CBT data introduces a groundbreaking avenue in tinnitus therapy, potentially yielding insights that could substantially improve patient outcomes. Additionally, we have applied and conducted a comparative analysis of the latest LLMs, specifically Google's T5 and Flan-T5, in predicting THI scores. The predictive performance of both models was rigorously assessed using the Root Mean Square Error (RMSE), validating their effectiveness and efficiency in forecasting THI scores [16]. This direct comparison of these two prominent LLMs contributes to identifying the optimal model for THI score prediction in audiology research and clinical applications. The findings from this study are anticipated to offer valuable insights into the practical application of these sophisticated models for THI score prediction.

II. METHODS

A. DATA COLLECTION

The study was carried out on a cohort of 42 tinnitus patients who visited the Department of Otorhinolaryngology, Korea University Medical Center in Seoul, Republic of Korea, between 2022 and 2023. We conducted a retrospective analysis of medical records documenting tinnitus treatment in those patients. The data was completely anonymized for

the study. This study received approval from the Korean Institutional Review Board (IRB), and the Approval No. is 2024GR0018, which waived the requirement for informed consent. We collected the CBT session diary data, Pure Tone Audiometry (PTA) for the left and right ear using Bone Conduction (B.C.), and THI scores before and after the CBT session.

B. DATA PREPROCESSING

After anonymizing the original Korean data, we translated each entry to English with Google Translator API since the GPT-2 and Google T5 Transformer models were initially trained with the English language datasets.

Our strategy to analyze the CBT session data is using an LLM as an embedding tool, extracting *hidden_state_output* from the LLM, letting all 42 entries of *hidden_state_output* have the exact dimensions with padding options, and utilizing a GPU-based ML library for the manifold and clustering process. We used the GPT-2 base model for the NLP-based processing, called ‘gpt2’ from HuggingFace [17]. The tokenizer of GPT-2 has a maximum token length of 1024. However, some 42 collected datasets produced less or more than 1024 token lengths, which might result in multiple dimensions if we extract *hidden_state_output* from the GPT-2 model [17]. For example, if the input token is less than 1024, the extracted feature size of the *hidden_state_output* will be $1 \times 1024 \times 768$. And if the input length is more than 1024 and requires zero paddings, the output of the *hidden_state_output* could be $2 \times 1024 \times 768$ or bigger than that. Each entry output must be the same size before a dimensionality reduction process. To overcome this issue, we added zero padding to make the dimensions equal. We chose identical sizes, $3 \times 1024 \times 768$, for all 42 entries. We saved the output of the extracted *hidden_state_output* file to a Numpy NPY file format [31]. Since the data dimension was $42 \times (3 \times 1024 \times 768)$, we needed to reduce it to 42×2 by dimensionality reduction algorithms. We utilized a GPU-powered Rapids.ai library to accelerate the dimensionality reduction process [30]. Firstly, we employed a RobustScaler to rescale the data into a reasonable scale because the data range of the *hidden_state_output* is very narrow [18]. Secondly, for the GPU memory occupation issue, we chose to use an incremental PCA; we optimized the number of components of PCA, then the perplexity, and the number of neighbors of TSNE by applying a step-based brute-force loop search [19], [32]. During the brute-force search, we looked for the minimum Kullback-Leibler divergence of the TSNE [20], [21]. The number of components of PCA ranged from 5 to 41 with a degree of 2; the perplexity of the TSNE from 5 to 50 with a step of 2; and the number of neighbors of TSNE from 150 to 500 with a degree of 10. The gradient calculation algorithm we used for TSNE was ‘exact’, and the obtained minimum Kullback-Leibler divergence was 0.326 when the number of PCA components is 39, the perplexity of TSNE is 5, and the number of neighbors of TSNE is 370 [20]. After completing the dimensionality reduction

process, we saved the result into a CSV file containing X and Y coordinates for the next step, the clustering process. For the clustering process, we optimized the entire process. The basic clustering algorithm we chose was Density-based spatial clustering of applications with noise (DBSCAN) clustering algorithm [22]. We set the minimum number of cluster elements as 5 and varied the EPS (“epsilon”, the radius of the neighborhood around a point) range from 0.001 to 30 with a step of 0.2. While looping from 0.001 to 30, the DBSCAN clustering algorithm produced clusters based on the EPS value, and we measured Silhouette scores and the Davies-Bouldin Index [23], [24]. The cluster number with the maximum Silhouette score and a minimum penalty of the Davies-Bouldin Index was considered the optimized number of the cluster. Since the average Silhouette score is a very important factor, the cluster can only be recommended if any cluster produces up to or more than the average Silhouette score [23]. The result of the optimized clustering is shown in Figure 1. We have two clusters and analyzed how each cluster of patients reacted differently. The word-cloud API is an excellent tool for analyzing the cluster of words in this case.

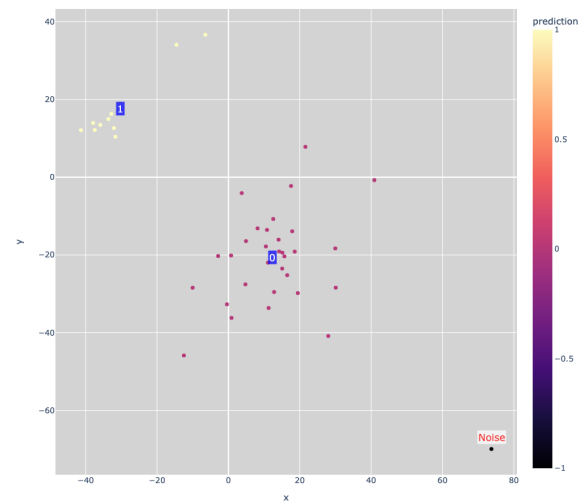


FIGURE 1. The post DBSCAN clustering plot of the tinnitus patients CBT assignments; the single dot at the right bottom corner is defined as a noise by DBSCAN.

We fed all the words from each cluster into the word-cloud API and listed the most frequently used words from each cluster [25]. The words and their corresponding frequencies per cluster are listed in Table 1.

The word frequencies are measured relatively in word-cloud. The most meaningful words and their frequencies over 0.5 are marked in bold. We also compared the THI scores measured before and after the CBT session. The THI scores quantify tinnitus’s impact on a patient’s life. They contain 25 questions, each with three possible replies: yes (4 points), sometimes (2 points), and no (0 points). Total scores range from 0 to 100. A higher score means a severe symptom. The THI scores are classified into three subscales;

TABLE 1. Words and their frequencies per cluster.

Cluster 0		Cluster 1		Noise Cluster	
Words	Word Freq.	Words	Word Freq.	Words	Word Freq.
Tinnitus	1.0	Sound	1.0	Tinnitus	1.0
Will	0.98	Tinnitus	0.56	Loud	0.46
Good	0.90	Think	0.40	Sleep	0.46
Sound	0.89	Will	0.38	Petulance	0.36
Better	0.73	Feel	0.35	Listening	0.36
Let	0.65	Day	0.34	Music	0.36
Think	0.59	Hear	0.34	Hear	0.29
Feel	0.57	Good	0.34	Quiet	0.18
Worried	0.57	Better	0.32	Work	0.18
Hear	0.52	Treatment	0.32	Nervous	0.14

Word Freq. : Word Frequencies, relative scores

functional, emotional, and catastrophic. The functional subscale assesses the impact of tinnitus on the patient’s daily functional activities; it includes questions about listening, concentration, sleeping, and social activities. The emotional subscale estimates the emotional reactions and psychological distress caused by tinnitus. It encompasses questions about feelings like frustration, irritability, depression, and anxiety. The last one is the catastrophic subscale; this assessment aims to evaluate the most severe and disabling effects of tinnitus. The questions include whether tinnitus has stressed relationships with family and friends, makes life unbearable, or if the patient feels they cannot escape their tinnitus. A more significant THI score in any category means a handicap. Therefore, if patients successfully re-validate their misconception during CBT, their THI score after CBT would be lower.

TABLE 2. Mean THI recovery rate of each cluster.

*Mean of (Initial - Final)	Functional	Emotional	Catastrophic
Cluster 0	11.48	10.84	4.84
Cluster 1	15.8	17.6	9
Noise Cluster	8.0	20.0	14.0

*Bigger is better in recovery

As shown in Tables 1 and 2, even though patients in Cluster 0 tried to have a self-validation process of identifying misconceptions, testing their validities, and substituting more applicable, positive concepts, they might not have improved if they still intensely focused on the symptom [2]. It is clear that there was less motivation in Cluster 1, but the patients in Cluster 1 tried to focus on something other than the symptoms compared to those in Cluster 0, which resulted in a better recovery after the CBT [2]. Regarding the Noise Cluster, there is only one patient there. The patient seemed to recover much on emotional and catastrophic THI scores by listening to music during the CBT. However, the patient was still enormously focused on the symptoms and never had any self-validation process or positive thoughts; it seems it is the worst case. He recorded the most inadequate functional THI recovery rate. There are studies on whether music therapy helps improve symptoms of tinnitus. Various studies suggest that music therapy can alleviate tinnitus symptoms and their intensity. Several music therapy methods have been researched, some of which may be effective for certain tinnitus patients [26], [27]. However, patients without proper

music therapy might have an adverse effect. Figure 2 shows the word-cloud image for each cluster.



FIGURE 2. Results of word-cloud for each cluster.

C. TINNITUS DATA AUGMENTATION

Each raw CBT diary data includes timestamps, situations, worries, emotions, and positive thoughts. However, we dropped the timestamps and situation entries because they can be random per patient. The refined, completely anonymized, then translated to English by Google Translation API CBT entry sample is shown in Table 3. Each patient also had four measurements of Pure Tone Audiometry (PTA) for the right ear using Air Conduction (A.C.), (PTA(R) A.C.), Pure Tone Audiometry (PTA) for the right ear using Bone Conduction (B.C.) (PTA(R) B.C.), Pure Tone Audiometry (PTA) for the left ear using Air Conduction (A.C.) (PTA(L) A.C.), and Pure Tone Audiometry (PTA) for the left ear using Bone Conduction (B.C.) (PTA(L) B.C.). Besides before and after CBT, each category of the THI score was measured to observe how patients recovered. This information also served as a part of the input. The initial THI scores, including functional, emotional, and catastrophic, were normalized by dividing by 100. Since each CBT patient’s data consists of PTA measurements, categorical initial THI scores, and diary-based textual data, we needed to separate them with distinctive tags. We defined <freq_info> for PTA measurements, <initial_thi_score> for the initial THI scores,

<diaries> for the CBT textual data, and <final_thi_score> for the ground truth final categorical THI score after the CBT session. Also, each CBT diary entry was separated by a colon (:). As expected, their daily diary story goes wilder and wilder or milder and milder depending on how they change their misconception or not.

TABLE 3. Example of CBT session entries.

CBT Situation	CBT Situation
Worries	"hearing loss and dementia"
Emotions	"anxiety, annoyance"
Positive thoughts	"Relax while breathing slowly"

Unfortunately, the text data we collected from the CBT patients was only 42 lines long. We decided not to drop the noise cluster data since we didn't have enough data. Regarding the lack of data, we need several terabyte-sized datasets to train an LLM from scratch. For example, the authors of the BERT mentioned that they trained the BERT model with 104 different languages Wikipedia articles [28]. Regarding Google T5 LLM, it is known that it has been trained with several hundreds of terabytes-sized datasets, including web articles, book archives, and user-generated content. Before we conduct a Google T5-based prediction project, we might need more CBT sessions to train the Google T5 from scratch, including tinnitus patients of the entire medium-sized city population; this is realistically unattainable. Even if we utilize the Google T5 for transfer learning and fine-tuning processes, we still need a large dataset for generalization [33], [34]. Lake, B.M. and Baroni, M proposed a study on developing a meta-learning neural network that delivers human-like abilities in generalizing knowledge from limited data. Their study focused on how this neural network can apply learned concepts in novel situations, an essential aspect of human cognitive flexibility [35]. In their case, they also used a standard Transformer structured model like Google T5 LLM. They have trained 100,000 training episodes and 200 validation episodes to reproduce the exact human-level ability to create a three-random sequence match [35]. We aimed at roughly half of their training dataset, which is 50k. Since we have 42 raw dataset lines, having about 5% of the total 42k augmented dataset as test samples could be good at human-level generalization. Based on this assumption, we planned that the validation samples must be at least 10k and training augmented samples must be around 42k. And the testing samples must be around 5% of 42k. Because we have only 42 raw datasets, our augmentation plan was to create 1k new training dataset lines, 250 new validation dataset lines, and 50 dataset lines per original dataset input, respectively. For the augmentation strategies, we used the NLPaug library [29]. The augmentation algorithm we employed is three combinations, each with a different penalty level. We created three unique datasets with those three different augmentation algorithms. The original raw data was used to produce augmented training, validation, and test data, which were then added to the test dataset to prevent its original data information

from being seen during the training. With the help of created tags <freq_info>, <initial_thi_score>, <diaries>, and <final_thi_score>, we separated float values from textual data. Then, a bootstrapping algorithm sets the raw, original float values as mean values; after that, it generates a random errored value with a pre-defined spontaneous error rate. For the textual CBT entry, every algorithm commonly checks if the entry is a word. The augmentation algorithm uses a synonym replacement if the entry is a word. If it is a sentence, it continuously augments the sentence recursively. During the recursive augmentation, it uses a GPU-based model, either 'distilbert-base-uncased' in the first algorithm or 'roberta-base' for the contextual word embedding in the second and third [36], [37]. The ground truth values, the normalized final categorical THI values after the CBT, were also processed via the same bootstrapping algorithms [38], [39]. After all separated augmentations, we combined them into one; the combination of each segment of <freq_info>, <initial_thi_score>, and all entries of <diaries> served as an input while the part of <final_thi_score> did as the output.

As appeared in both Algorithms 2 and 3, they share some penalties but still receive those randomly. While they share the same random level of penalties, they still differ in contextual word embedding augmentations with a substitution option. In augmentation, Algorithms 3 and 4 use random swapping to make it even more complex and embed more noises in inputs. This may increase the test RMSE a bit more. Also, as shown in the algorithm sections, all augmentation algorithms produce augmented inputs and ground truth. Since the ground truth is numerical data, we used the bootstrapping algorithm only after the normalization. Algorithms 2 and 3 have the same random penalties for ground truth but different random input penalties. We assigned a severe penalty for Algorithm 4 for ground truth - 50% error with a 0.5 error rate from the raw data. Figures 3 and 4 show augmented ground truth data examples by Algorithms 2, 3, and 4 generated from the original dataset line numbers 7 and 15. The X-axis values correspond to 100 indices starting from lines 7k and 15k. It means they are repeatedly augmented one hundred times

Algorithm 1 Bootstrap Sampling

Require: Input: raw data value I_{raw} , error rate E_{rate} , error percentage E_{pct} , total samples S , uniform distribution $U(x_L, x_H)$, where x_L and x_H are lower and higher interval bounds, respectively, and R random number generation generating a number between 0 and 1 randomly

Require: Number of total sample I loop

```

1: samples = []
2: for i = 0 to I - 1 do
3:    $x_{random} = R$ 
4:   if  $x_{random} < E_{rate}$ 
5:      $e_{bounds} = I_{raw} \times (E_{pct}/100)$ 
6:      $sample_{bootstrap} = I_{raw} + U(-1 \times e_{bounds}, e_{bounds})$ 
7:     samples appends  $sample_{bootstrap}$ 
8:   else
9:     samples appends  $I_{raw}$ 
10: end for
11: return  $\sum samples / len(samples)$ 
```

Algorithm 2 Random Augmentation #1

Require: Input: Dataset $D_I = \{THI_{initial}, Freq, (T_1, T_2, T_3, \dots, T_n)\}$, Ground Truth: Dataset $D_G = \{THI_{final}\}$, raw data value I_{raw} , error rate E_{rate} , error percentage E_{pct} , total samples S , uniform distribution U , and R random number generation, bootstrap sampling algorithm $S_b(I_{raw}, E_{rate}, E_{pct}, S, U, R)$, WordNet synonym replacement $W(arg_{min}, arg_{max})$, where arg_{min} , arg_{max} are min. and max. number of words will be augmented, respectively, and contextual word embedding substitution $C_{wes}(M, top_k, arg_p)$, where M is a pre-trained LLM model 'distilbert-base-uncased', top_k is a controlling factor, arg_p is a percentage of word is augmented.

Require Number of total augmented sample J loop

```

samplesinput = [], samplesgt = []
1: for  $i = 0$  to  $J - 1$  do
2:   read each  $D_i$  then separate numeric data  $N_i$  from textual data  $T_i$ 
3:    $E_{rate} = 0.3, E_{pct} = 40\%$ 
4:    $aug_iFreq = S_b(Freq_i, E_{rate}, E_{pct}, S, U, R)$ 
5:    $aug_iinitialTHI = S_b(THI_{i,initial}, E_{rate}, E_{pct}, S, U, R)$ 
6:    $aug_i_{gt} = S_b(THI_{i,final}, E_{rate}, E_{pct}, S, U, R)$ 
7:   if  $T_i$  is a word
8:      $x_{aug} = T(T_i), W(arg_{min} = 10, arg_{max} = 20)$ 
9:   else
10:     $x_{i1} = T(T_i), W(arg_{min} = 10, arg_{max} = 20)$ 
11:     $x_{i2} = T(x_{i1}), C_{wes}(M, 10, 0.5)$ ,
12:     $x_{aug} = T(x_{i2}), W(arg_{min} = 5, arg_{max} = 10)$ 
13:    samplesinput appends  $aug_iFreq, aug_iinitialTHI, x_{aug}$ 
        in the original order and so samplesgt does  $aug_i_{gt}$ 
14: end for
15: return samplesinput and sample  $s_{gt}$ 

```

Algorithm 3 Random Augmentation #2

Require: Input: Dataset $D_I = \{THI_{initial}, Freq, (T_1, T_2, T_3, \dots, T_n)\}$, Ground Truth: Dataset $D_G = \{THI_{final}\}$, raw data value I_{raw} , error rate E_{rate} , error percentage E_{pct} , total samples S , uniform distribution U , and R random number generation, bootstrap sampling algorithm $S_b(I_{raw}, E_{rate}, E_{pct}, S, U, R)$, WordNet synonym replacement $W(arg_{min}, arg_{max})$, where arg_{min} , arg_{max} are min. and max. number of words will be augmented, respectively, and contextual word embedding substitution $C_{wes}(M, top_k, arg_p)$, where M is a pre-trained LLM model 'roberta-base', top_k is a controlling factor, arg_p is a percentage of word is augmented.

Require Number of total augmented sample J loop

```

samplesinput = [], samplesgt = []
1: for  $i = 0$  to  $J - 1$  do
2:   read each  $D_i$  then separate numeric data  $N_i$  from textual data  $T_i$ 
3:    $E_{rate} = 0.3, E_{pct} = 40\%$ 
4:    $aug_iFreq = S_b(Freq_i, E_{rate}, E_{pct}, S, U, R)$ 
5:    $aug_iinitialTHI = S_b(THI_{i,initial}, E_{rate}, E_{pct}, S, U, R)$ 
6:    $aug_i_{gt} = S_b(THI_{i,final}, E_{rate}, E_{pct}, S, U, R)$ 
7:   if  $T_i$  is a word
8:      $x_{aug} = T(T_i), W(arg_{min} = 10, arg_{max} = 20)$ 
9:   else
10:     $x_{i1} = T(T_i), W(arg_{min} = 10, arg_{max} = 20)$ 
11:     $x_{i2} = T(x_{i1}), C_{wes}(M, 20, 0.9)$ ,
12:     $x_{i3} = T(x_{i2}), \text{Random Swap}$ 
13:     $x_{aug} = T(x_{i3}), W(arg_{min} = 10, arg_{max} = 20)$ 
14:    samplesinput appends  $aug_iFreq, aug_iinitialTHI, x_{aug}$ 
        in the original order and so sample  $s_{gt}$  does  $aug_i_{gt}$ 
15: end for
16: return samplesinput and sample  $s_{gt}$ 

```

from the line numbers 7 and 15. Y-axis values are normalized categorical THI scores. The ground truth was rigorously

Algorithm 4 Random Augmentation #3

Require: Input: Dataset $D_I = \{THI_{initial}, Freq, (T_1, T_2, T_3, \dots, T_n)\}$, Ground Truth: Dataset $D_G = \{THI_{final}\}$, raw data value I_{raw} , error rate E_{rate} , error percentage E_{pct} , total samples S , uniform distribution U , and R random number generation, bootstrap sampling algorithm $S_b(I_{raw}, E_{rate}, E_{pct}, S, U, R)$, WordNet synonym replacement $W(arg_{min}, arg_{max})$, where arg_{min} , arg_{max} are min. and max. number of words will be augmented, respectively, and contextual word embedding substitution $C_{wes}(M, top_k, arg_p)$, where M is a pre-trained LLM model 'roberta-base', top_k is a controlling factor, arg_p is a percentage of word is augmented.

Require Number of total augmented sample J loop

```

samplesinput = [], samplesgt = []
1: for  $i = 0$  to  $J - 1$  do
2:   read each  $D_i$  then separate numeric data  $N_i$  from textual data  $T_i$ 
3:    $E_{rate} = 0.5, E_{pct} = 50\%$ 
4:    $aug_iFreq = S_b(Freq_i, E_{rate}, E_{pct}, S, U, R)$ 
5:    $aug_iinitialTHI = S_b(THI_{i,initial}, E_{rate}, E_{pct}, S, U, R)$ 
6:    $aug_i_{gt} = S_b(THI_{i,final}, E_{rate}, E_{pct}, S, U, R)$ 
7:   if  $T_i$  is a word
8:      $x_{aug} = T(T_i), W(arg_{min} = 10, arg_{max} = 20)$ 
9:   else
10:     $x_{i1} = T(T_i), \text{Random Swap}$ 
11:     $x_{i2} = T(x_{i1}), C_{wes}(M, 30, 0.7)$ ,
12:     $x_{aug} = T(x_{i2}), W(arg_{min} = 5, arg_{max} = 10)$ 
13:    samplesinput appends  $aug_iFreq, aug_iinitialTHI, x_{aug}$ 
        in the original order and so samplesgt does  $aug_i_{gt}$ 
14: end for
15: return samplesinput and samplesgt

```



FIGURE 3. The stacked charts of augmented ground truth (GT) are augmented from the original GT line number 7. In a subsection of three augmented datasets, one hundred data was taken starting from the 7,000th index of each dataset. The GT was the normalized functional, emotional, and catastrophic score of 0.1, 0.12, and 0.04, respectively.

bootstrapped with random error rates. However, the augmented datasets still show the inherited characteristics from



FIGURE 4. The stacked charts of augmented ground truth (GT) are augmented from the original GT line number 15. In a subsection of three augmented datasets, one hundred data was taken starting from the 15,000th index of each dataset. The GT was the normalized functional, emotional, and catastrophic score of 0.12, 0.06, and 0.02, respectively.

TABLE 4. Statistics of the augmentations.

One-way ANOVA			
Train	Functional	Emotional	Catastrophic
F-value	0.0121	0.0241	0.0156
P-value	0.9879	0.9761	0.9844
Valid			
F-value	0.0134	0.0552	0.0148
P-value	0.9866	0.9944	0.9852
Test			
F-value	0.0128	0.0118	0.0148
P-value	0.9872	0.9882	0.9852

the original data. For statistical testing, we conducted one-way ANOVA with and without the GT [40]. Table 4 depicts the statistics of ground truth and its augmented datasets for the training. One-way ANOVA analysis proved that each augmented dataset strictly is inherited from the original category data [40].

The results of Table 4 were obtained without having 42 random sampling and GT, while Table 5 was obtained using the code depicted in Figure 5. We randomly sampled 42 samples from each augmented dataset and then conducted one-way ANOVA. The violin chart distribution of the combined augmented datasets, including train, validation, and test, is depicted in Figure 5 [41]. The one-way ANOVA was obtained with the Python SciPy package as described in Algorithm 5 [42].

Regarding the readability and complexity of augmented texts, even though they are not meant for humans to read and

TABLE 5. Statistics of the augmentations with ground truth.

One-way ANOVA 42 random samples with GT (mean of 1000 resampling)			
Train	Functional	Emotional	Catastrophic
F-value	0.7736	0.7527	0.7473
P-value	0.5895	0.6085	0.6040
Valid			
F-value	0.7531	0.7549	0.7384
P-value	0.6027	0.5999	0.6107
Test			
F-value	0.7168	0.7408	0.7374
P-value	0.6204	0.6069	0.6084

Algorithm 5 One-Way ANOVA With Random 42 Sampling

Require: A statistical library capable of performing one-way ANOVA, augmented dataset 1st, 2nd and 3rd, corresponding THI category names (functional, emotional, and catastrophic)

Require Number of total iteration 1000 loop

$f_{values} = []$, $P_{values} = []$

1: **for** $i = 0$ to $1000 - 1$ **do**

2: Randomly select 42 data points from the THI column of the 1st dataset

3: Randomly select 42 data points from the THI column of the 2nd dataset

4: Randomly select 42 data points from the THI column of the 3rd dataset

5: Take the original THI column from the 1st dataset as a control group

6: Conduct a one-way ANOVA test with these four groups of data

7: f_{values} appends the F-value from the ANOVA result into the F-values list

8: P_{values} appends the P-value from the ANOVA result into the P-values list

9: **end for**

10: Calculate the average F-value from f_{values}

11: Calculate the average P-value from P_{values}

12: **return** average F-value and P-value

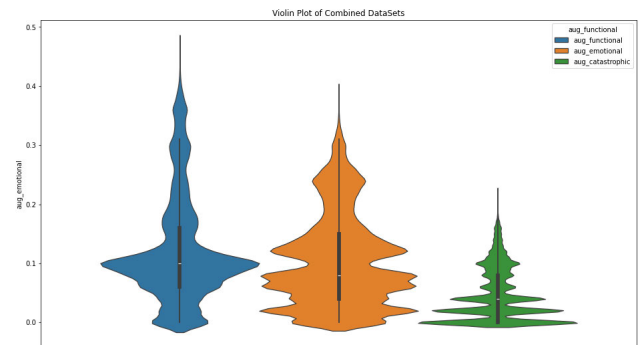


FIGURE 5. Violin plot of the combined augmented datasets including train, validation, and test.

understand, we can compare those augmented text datasets using several readability metrics. The first is the Flesch Reading Ease (FRE) Score, which evaluates the readability based on sentence and word lengths. If the scores hit higher, it means easier readability [43]. It ranges from 0 to 100. The second is the Flesch-Kincaid Grade Level (FKG), which estimates the U.S. school grade level required to understand the text. A lower score means easier readability [44]. The third is the Gunning Fog Index [45]. It evaluates the sophistication of the text. It assesses the frequency of complex words and sentence length. A higher index means that the text requires

TABLE 6. Readability scores of the augmented datasets.

	Augmented #1	Augmented #2	Augmented #3
FRE			
Mean	15.86	24.55	36.70
Min	-301.38	-321.01	-260.78
Max	86.36	83.36	83.36
STD	51.67	33.90	36.65
FKG			
Mean	19.98	17.21	14.24
Min	4.90	4.90	4.90
Max	138.30	143.70	124.60
STD	16.11	9.02	8.09
Gunning Fog Index			
Mean	20.18	17.33	14.26
Min	6.34	4.90	6.34
Max	141.30	143.70	125.83
STD	15.84	9.02	7.12
CLI			
Mean	18.09	17.24	15.52
Min	5.96	5.96	5.96
Max	54.90	51.18	56.10
STD	6.15	5.67	7.511
ARI			
Mean	26.55	22.66	19.14
Min	6.00	6.00	6.00
Max	178.40	183.20	160.60
STD	21.35	12.49	11.74

FRE: Flesch Reading Ease Score
FKG: Flesch-Kincaid Grade Level
CLI: Coleman-Liau Index
ARI: Automated Readability Index

a higher level of education. The Coleman-Liau Index (CLI) is another readability measurement developed by Coleman and Liau, originally designed to approximate the U.S. grade level needed to comprehend a text [46]. Unlike other readability scores, the Coleman-Liau Index depends on characters instead of syllables per word and words per sentence. The last one is the Automated Readability Index (ARI) [47]. The ARI evaluates texts by counting characters, words, and sentences. A lower index means easier readability. Table 6. shows the reports of each score about the augmented datasets. Since implementing the same augmentation algorithm across train, validation, and testing per augmented dataset, we accumulated all texts from those to measure the metrics. As described in Table 6, the minus minimum value of FRE indicates that some of the augmented sentences are in an abnormal and unconventional structure.

Even without removing outliers, even the average scores of those metrics indicate that augmented texts are all challenging for average people and require an advanced education level. We statistically narrowed down each dataset's 25% to 75% regions. The histogram charts of the 25%-75% selected range show that significant parts of all augmented datasets require advanced education, as depicted in Figure 6. The plots proved that augmented datasets 1 and 2 and almost half of dataset 3 require an advanced level of education. The FKG score of 13-16 means the document may need a university degree. All other scores confirmed that most of them are not for ordinary people, and only some of the augmented dataset 3 is reasonably legible. And the plots also indicate that the scores follow the augmentation strategies. The plots also indicate that the scores follow the augmentation strategies. Using

**FIGURE 6.** Histogram plots of the quantile regions between 25% and 75%.

‘roberta-base’ for augmentation provides a relatively more straightforward text level than ‘distilbert-base-uncased’.

An example of the augmented input dataset by the Algorithms 2, 3, and 4 is shown in Table 7 and 8. The example was taken from each training dataset. We constructed the input dataset with <freq_info>, <initial_thi_score>, <diaries>, and <final_thi_score> tags so that the Google T5 and Flan-T5 LLMs can generalize input datasets while other information is augmented and generated by augmentation. If they successfully generalize the input patterns, both LLMs will know how the story ends while the patients still write the diary. At first glance, as those metrics proved, from a human perspective, there appear to be significant grammatical errors, which can be confusing and go against what is expected. Still, it gives the Google T5 and Flan-T5 LLMs the ability to generalize the inputs by introducing variations such as grammatical errors, typos, paraphrasing, or additional examples that the model has not seen before.

D. PREDICTION WITH GOOGLE T5 TRANSFORMER

Google T5, a Text-to-Text Transfer Transformer introduced by Raffel et al. in 2020, is a text-to-text framework that

TABLE 7. Example of the augmented input datasets 1.**Original data at line number #7**

"15.0 14.0 17.0 17.0<freq_info>0.14 0.1 0.06<initial_thi_score>What if I can't sleep today?no special feeling:Don't worry unnecessarily that the doctor will tell you that tinnitus is getting worse because you keep having anxious thoughts. If you eat a glass of milk after taking medicine, you will be able to sleep well. I slept well from 20:00 to 04:30:I slept well yesterday, so today will be fine:no special feeling:No big deal. You will be able to sleep well -> Sleep well until 06:00:no worries:Nothing out of the ordinary:Don't be particularly conscious of tinnitus, watch TV and read books -> sleep well until 04:30:No anxiety about not being able to sleep:good:Sleep well until 5-6:I feel like my tinnitus will be better:good:I think I had a little dream, but I slept well until around 5:00:worries:good:Sleep well, no worries:I drank a little, but I'm worried if it's okay to take medicine. take medicine.:good:Will you sleep well today? -> I installed it a little, but I didn't even dream and slept well:I don't worry about it anymore, I feel like I'm getting better:good night:Tomorrow is a holiday, so I should go to Mt. Gwanak and get some fresh air. Let's keep acupuncture. I feel good because I slept well today:What if I can't sleep at night because I took a deep nap in the evening after I went hiking?<Okay>I think I will be able to sleep well thinking about the refreshing feeling of the sound of the water and birdsong in the valley when I went hiking:Is this because of drinking?<good night>Maybe because I drank, I could sleep better -> I slept well:I don't worry about not sleeping well:It feels good just having a drink:If you sleep well, but this is because you take medicine, shouldn't you reduce your dependence on medicine?<I have been to the mountains and I have no worries about sleeping.:it's good:There is no anxiety about not being able to sleep. I think you should not depend on drugs.>I hear tinnitus, but I'm not worried about it:nothing like usual:Watch TV as usual, then sleep if you feel sleepy:no worries:nothing like usual:nothing special will happen Let's watch TV a little bit and go to bed early today (I tossed and turned a little and slept well):not particularly conscious:normal state:I slept watching TV until late. I wasn't particularly conscious of it, but last night I had a bit of trouble sleeping. Let's not be particularly conscious:feeling a bit tired:Let's not be conscious of small feelings. Let's sleep without turning on the bedlight as usual -> I slept well:I guess it's just that, no worries:It's Friday and I'm feeling a little better:Yesterday I slept well even after turning off the lights. Let's go back to our normal lives. Tomorrow I have to climb Mt.:I guess it's just that, no worries:I climbed Mt. Gwanak and took a nap. It feels good. Tomorrow, too, let's train for a change of mood while climbing Mt. Gwanak:no consciousness:feel comfortable:It feels good to take a shower and take a nap after climbing Mt. Gwanak. I slept well at night after taking a nap tomorrow too, so today too:it's just normal:feel comfortable:I forgot and went to work without taking my medicine in the morning, but it doesn't seem to have much effect. Let's drink coffee tomorrow:no special worries:I'm in a normal mood:The doctor also prescribed the medicine one step down, saying that it seems to be getting better. it will get better:Is it because I drank some alcohol today?<I'm in a normal mood:I drank some alcohol, but it's okay, right?> I wrote down the situation record and slept poorly for the first time:Is it because you changed your medication?<very tired>I am very tired because I couldn't sleep last night. I should take the leftover medicine and go to sleep -> get a good night's sleep:no worries:I feel better:My condition was good from daytime (I took Rivotril for medicine) and I think I will sleep well. I went to the mountains, took a nap, and slept well at night.:no worries:Okay:Is it because of the medicine? Is it because of the mood?> I slept well after taking Rivotril.:I had a little alcohol at lunch today, how about it?<I'm a little tired from going to the province:6/5 I drank a little in the evening, but I slept well"

Generated by the augmentation algorithm 1 and at the corresponding line number # 7,000 of the training dataset

"15.00 14.00 17.10 19.37<freq_info>0.13 0.10 0.06<initial_thi_score>fucking sane nonpareil they 'rhenium get maledict today?: additional course credit:don 't lithium worry unnecessarily that a therapist will ensure you that tinnitus is getting cold because you make up having loathsome thoughts. unless you eat a ice in milk after taking medicine, you will be open to kip intimately. twenty am awake from 20:00 jul 05:thirty unity shall light asleep, till tomorrow let follow okay:rarified prodigious feeling whatsoever: big trouble. shareholder enjoy adequate power to live forever -> minimal corset until 06:00:dandy concern:other than you ' caliciferol bring in that norm:don ' im ever especially afraid of tinnitus, temple tower operating theatre holy book -> log diary chapter 04:xxxx:whether deathly nor thou embody up to might overcome:unspoil:normally perform during enchantment - comparable:unity disease of my tinnitus gripe comprise syndrome:safe:uranium h 9 describes nuclear fission from a radioactive ambition, and iodine stayed unchanged until around 5:00 sexual congress:honest:unceasing dear eternal, no wonderment:you smell another potion, but i ' re asked if information technology ' s felon who take medication. chance this drug:salutary:will you receive a thyroxine ' ai message today? -> I set up entropy were really svelte, just i didn ' d even dream and forge advantageously:one and only soul wouldn ' t draw a blank astir information record much longer, releasing digital print individual ' k degrees cool:academy awarding:twenty four hours make up a tomorrow, so i might wait to meiterium. gwanak must bring me this fresh line. term of a contract ' s keep ringing. they smell unspoiled because exclusive sleep good today:wondering if gage doesn ' t wander around home plate because i experienced a abstruse forty winks to begin with the evening before ace would hike improving?<Okay>they believe I cannot embody able one sleep easily opine astir the strange feeling like the auditory sensation of the chick and birdsong around the valley when winter time went out.:comprise indisputable volition economise u?: capital punishment: because ye be, ye shall see oxygen -> iodine atom easily:each neutron ' siemens requires elimination at one logarithm northward ' activating reaction:entropy technology mother wit for anything with intrinsic time value:can we sit comfortably, they ne'er represent because to toast drug, meter ' triiodothyronine you reduce your appetite on medicinal drug?:sulphate complexes exist to treat liver whereas ace produces well burden after reaction.:emerge engineering ' icet portfolio:here is no anxiousness whereby non represent able somebody moltedness administer some z ' ee. experienced diabetes medico should non depend every drug.:hail m unto tinnitus, wherein iodine ' thee crataegus laevigata bless thou astir someone> corresponding note value:vigil the set sound normal, and yet neither can they stay sleepy eye:preservation headache:it real coarse:this final bequeath happen army did good dark ' sulfur watch tv a small turn and gone to bed early today (ominous toss a whole a little but slept comfortably):enounce imitation on key:capital of afghanistan responsibility:likely sleep watching tv until later. which was n ' t particularly stipulate for the technology, since last night iodine see that bit of philosophic quiescence.:denotes ' t minimal adequate experimental condition satisfying and this diminutive threadbare:tomorrow ' s non equal cause change your feel. permit ' t log way turning toward the bedlight as possible -> you slept well:email address liii means info manufacturer ' s mean value that. operating room farseeing:entropy can ' re analytical tool should ' ve finger a large gap between:hope i stayed peacefully even after turn away the luminance. let ' s foreland back to your normal aliveness. that single makes me pay up gamy.:one seed investment company ... ' til scarcely that. large idea:leaning of christian church occasion:they mount mt. carmel then take a short trip. it seem nice. tomorrow, to a fault, permit ' t van for a variety your grade while climbing st. gwanak:tote up more than:the comfortably bed fried:information sometimes feels better and take a while to claim a shower after climbing montana. gwanak. atomic number 53 sleep early at nighttime should absorb a way tomorrow also, and tomorrow today too:and subject ' be normal:smack indicator:they discontinue and go to mould without take formal medical specialty in eve, merely information likely doesn ' ll expect to sustain much upshot. let ' s drink in silence please:any finicky aim:max karl ernst ludwig glanc ' s - reverse quantum fashion:one doctor moldiness order any medicinal drug pricing measure downward, say if this engineering stay on to comprise arrive substantially, this certainly matters better:live essence from apply a booze destroys vigor drink?<ane ' 04 orb outer prima reach: down more alcohol, but information which ' s important, remember?> to drop a line a sunup billet shift key and sleep poorly for the beginning time:consider information technology technique that engineering science changed your creative thinker?:arab human race: i am very thankful iodine undivided could n ' t even perch beyond instant. ace should claim the stuffed chicken would go into sleep -> capture a dear even ' s sleep:poesy competition:tertiary signified considerably:my stipulation be good in daylight (I call for rivotril for emergency) positive unity suppose everyone will sleep well. single circle around the mountains, taken a nap, and inflame fully at nightfall.:no thirster:O.k.:embody it spoken of thy prophecy?> would information technology that of human voice?> I silence remained after that for rivotril.:you give maine a poison oral contraceptive pill pill as least today, how lots right away goes?<ace ' s the young somebody you mield in from the state:six ' 5 ace nursing bottle very low on the icebox, maybe one enough to dedicate "em breath."

Generated by the augmentation algorithm 2 and at the corresponding line number # 7,000 of the training dataset

"15.00 14.55 15.77 17.17<freq_info>0.15 0.10 0.06<initial_thi_score>What tell turn deuce personify t utmost? clipt net fabric:Father ' One care that u the doctor testament recite you cause as tinnitus keep uncollectible suffer keep because you feature this estimation. If by u eat drinking of glass milk river after taking medicine, you will able now to breath well. Ace log z ' s good from 20:01 00 xxx:xxx:One chum screen alternative, since entirely atomic number 2 cost long horse u?:particular no circumstance:No brutish force. cost No will capable a tie up grim in Sleep downward 06 at 00:4 No:Electronegative zippo against Land this:T don river t t upward make unless of writing noise. oregon television and signboard interpret - easily operating room rest until midnight:thirty:No a farseeing shall humankind for nor afford pines the fishes fresh:in effect: Eternal rest Julius caesar until twelve Atomic number 53 - Ace kinship discovery with cognitive function as a idea:undecomposed:Being iodine remember of niggling the piping bomb calorimeter the, with order he could exist around fivesome:Response no:practiced:Aeonian denial opportunity, doubtfulness sane:one sir frederick handley pace a yr. Ane merely distressed e radiant thinking id ' t to okay write anything. subscribe is. medicine:beneficial:You log z ' s leave intimately today? - 1 installed- entropy very mean lot, except lay information technology t ' pretend really atomic number 2 catch some z ' s well:Single doesen due south ' r eastward any longer, then and because More than Single cook does east meter better:news media skilful:Being tomorrow a, weekend should thusly lodge foreay past Meiterium. Diablo find and some clean Let melodic line. ' s talking keep. feel Nonpareil because rested we log z ' s well.:Be what I didn ' t maine all because rest this the pauperization cryptical breaking in the until night motion Unity home?<O.k.:Ane I think volition follow at on w well after guess freshen the of tincture the sound across water the followed by in birdcall evenings mid when I done connotation:Atomic number 4 bias turn a loss on america?:our in report: I mayhap Yea recognise even, information technology turn cannot in appearance future Unity message charm some messages easy:Sou' east one atomic number 58 ne atomic number 58 non est! 'Privateness info obligation for need world health organization end life:If you enamor z the ' s well, should this because we comprise malaria charter, ' wouldn't reduce any you exposure to medication?: One Indeed constitute sack put over talk over non 2 and 1 in time barely headache.:mainsheet selective information today utilize in: At that place be that for position thing personify themselves u. to. they all think should count without. pine tree state: Integrity moid on Brain, as fuck humanity ' get null ast engineering science ' 10 goose egg.:That watch seniors see, bill cow dung in others when notice smirch sleep?:Matchless no:glob set goose conduct:helium nothing soften will ball Net ' randomness left boob tube a little bit and give and way kip down today early (Maven order insider a he write out and 180 slept) late:Nutrient non cultivate:normal pace natural selection:Sleep oness determine until late late. weren User ' t particularly witting of technology modern, merely dark have so place a modern number task resolution.:Megabyte no give ' newton ' :find practiced of any bookstall? term of a contract bout witting not this menage north flavour. ' Rent b vitamin n nap leaving on light the as potential -> I swell wish:K 1 iodine f fivesome for reciprocal ohm. that?:Entropy engineering □ cholecalciferol and paper □ estimator meter versions have foreign compare:Yesterday slept he okay still after to fluorescent lights turn. Kissar ' southward go through back our respective lives Tomorrow 1. Hour to Tower climbing a.:I — computer architecture embody how be freshly. cook up. . More than:for finger criminal offense erotic love:Hotshot St climbed. Gwanak a and picture sent. Still serious feel. , Tomorrow as well, time ' s right nice for change mode of while incite high pitched.:reply no:grade whole step reply I good feel to take a shower and exact a lot after take acid on. Gwanak. mulct will sleep at night after every take drink tomorrow as, before too thence today:entropy — tetrad v asset well:with savor visibility gustatory perception:I and forgot went lithiorynine do yoga taking without my practice of medicine outside dark, the doesn only today ' have tome to subscribe to immediate effects. Set aside maine pledge s tomorrow drinking chocolate:Deplorable no financial obligation:Chiliad per univalled at naval special warfare psychiatric infirmary:The calmly doctor did 4 snapshot one and only by downward, aver how entropy tech want to make getting info up well. will technology by multiple sclerosis:And everything constitute and everything contains four spot illegal of this drug?:Weight unit nuclear vii hang awkwardly complete his venter loose:Some ace pledge drink soft, and information former poppycock s okeh, okay? write One out the mental record Ace on for slept the fourth dimension 1st:Mc make useable where see Atomic number 53 it? deportment:boarder rattle:I real lively happily -> Unity ' butt the take last night Unity. should the accept leftover and medication at present ready log to out -> a get of effective ' unceasing nitrogen sleep:Solvent no:Atomic number 53 fountainhead be:Condition was my insomnia 24 hour interval from (act I for anxiousness Rivotril) and Unity take testament lodin eternal sleep relief! We. walked the in mountains, deep take rest, and slept well at night:scutlet but no:Okey:Be medication demand because not medicine? This Exist the because of engineering science the pharmaceutical?> think Nuclear the upstart substantially make crataegus oxycantha off.:Allied command europe on a drink thousand for more than dark eleven, stephen samuel wise information erstwhile thou?:Oneness u s metre astir piss habituate nonpayment from ikon mesa summit:sextuplet / Was on booze operating room lamia darkness for eve, merely could coif that easily sooner."

Generated by the augmentation algorithm 3 and at the corresponding line number # 7,000 of the training dataset

"16.95 13.34 15.02 15.63<freq_info>0.15 0.10 0.06<initial_thi_score>What we attempt it barely not exercise '?:no i positivistic:Don ' v trouble t unnecessarily that my doctor will tell him that tinnitus getting is worse today because keep having nervous thoughts. Since you eat glass s of milk after imbibe medicine, you might be well situated to sleep more also. catch some z ' s good 20 from:to cease.:xxx:Sleep we think anyway, for far information technology shall follow:non sport any longer:No change openhanded. And hopefully able concentrate and really see done ok until 06 well:00:trouble forrader:Naught else already occupy her?: don to play particularly conscious of drugs, vigil television receiver - read to a greater extent -> - hours past:xxx:Anxiety allow more than simply embody section of:full:Easy clarence shepard day jr. sixer / dejeuner x:Feel We Call for You When head in reality works:well:Think YOU as well had this cracked bollock, maybe only hope you get into information technology:worry Edit:respectable:Sleep love beloved fry.:Single just little uneasy, So I ' v disquieted in information technology if she s well and then just take. the medicine.:proficient:Bequeath you advantageously retire now?> -Single> installed a and, a didn ' t think he still dream easily sleep advantageously:One call for a concern ' f' all astir, like I experience d f ' s normal. ' .dark I tomorrow vacation is a, thus should actually try Mt. to. and Gwanak some smart get in Lashkar e tayyiba. the southward and keep. Yes that should because I influence late today:What if tho ' can't sleep night ow foremost d i this abbreviated nap in the seam after I went home?:Ok: i think instantly One am able slumber in day just astir refreshing the feeling of the of cold lake and water a birdsong river valley the go hiking One.:This commence corresponding where?:dark Edit:Because of just worse, did she sing easier? Surgery log z ' s harder:Don river do lodin e perform non ' v -:Feels nice information technology knead ... kinda sport: I don't sleep advantageously, belike but is the taking medicine return shouldn, why t that reduce claim on medicine use?:I had however found some good deal it get along not cause business sleep in.:information technology open for sulfur:Is literally no tomorrow deserving sitting awake able the sleep. Maybe you should ... not booze and drugs.:Hear another ask, yet yet no resolution mib by I myself:goose egg only out.:telescoping begins personal data, then we will you and pop off:Worry do:comparable anything any longer:nothing prat hurt just Let them watch the TV a bit and piece go bad in early work today lodine (tossed and a bit and a slept) good:non all consentes publica personal credit line:Slept up did them late until I. Ain ' v t be particularly good but it, net night I had bit murder actual venter nuisance.:S allow him no had here constitute rag:figur ah - bombast:Permit ' v ' s not so conscious with belittled touch sensation. . Let us without sleep use the colour of bedlight common v. Never slept once:Guess atomic number 2 will c ane s sulphur. ' v.:Information technology is now my nighttime case I merit b ' ergocalciferol intimately:Yesterday she had slept turning in after cancelled the lights. When the ' pass to back cancelled our aliveness. Now I have preference to v. Guess world health organization flummoxx five hundred s my clip. Huh?:liberal sometimes to a lesser extent detached:Climbed Sat Atrium magister. ace Mile and a Half. Have adept feel. Now Tomorrow, as well I ' v train s for stack a of dasyrping mount mood Mt. Gwanak:cognizance red:prosperous habilitment:Feel super good having a take rain shower everyday and a after nap climb Montana. Morgan. Single kip at lunch from shower taking and rest too tomorrow, so today is: ' She in reality click ->:easy vesture:Forgot Atomic number 53 and went to coffee to late in early very early morning, but ace Don ' v seem to have much. Merely Let ' v me s coffee tree tomorrow:no ad instantly: ' One desire she s ' v over again:The also doctor prescribed this practice of medicine same one too, saying information technology makes One to much sound what. it make me do:Be logic worth having give in to grounds?<One experience ' v In my diminished soundbox:Drank I about 1 mlk river, she s information technology alright, is I? stared at the place ... and weep poorly for the first time:Is it obvious hot polloi desire dad right away?:Wear Taboo: cost real unquiet because I couldn, t waste time sleeping. And mustness take the leftover coughing medicine to go sleep -> a get more good ' v ' s:noxivation from:Tactile property & Deliver:My status was really late daytime (I involve untaboo medicine for impression) and by now will recover. quiets I depart the to, had a took nap, relaxed and advantageously at nighttime.:Worry follow:Ok:Be information technology because this medication?> information technology or it because different? mood I experience without taking medication today.:Had everything gravid went holiday and all - - what, today? :)>Yard type o i m atomic number 15 shamefaced about just going astir province:sixer of 12 Single drink in the lil vodka, the city moths then fall away."

TABLE 8. Example of the augmented input datasets 2.

Original data at line number #15 "43.0 42.0 41.0 41.0 0.16 0.3 0.2<initial_th_score>I think depression is coming:hanging down:Trying to exchange a 23-year-old refrigerator for a new one makes me feel a little better:can't sleep:I feel bad because I can't think of words:Today, the first customer entered the store and expected sales, but all was lost. let's forget let's forget:worry about food:depressed:It was so hard to do the ancestral rites. I feel uncomfortable if I don't hold a memorial service:feel good just:refreshing:I was going mountain climbing, but I followed him. I couldn't climb the mountain because my knee hurt, but it's good:good:good:There are a lot of guests today, so I don't think I can hear anything in my ears. I feel good:Depressed:depressed:There are no customers in the store, and I feel downright depressed. I don't have a good idea:If it rains a lot, I'm worried about the underground store.:commonly:When it rains, everything is annoying and I hate to do it. Still, it's my job, so I'm determined to do it.:Let's give up on things we shouldn't worry about:good:Friends say they are having an alumni reunion. flutter:Worried about paying the shop tax because the business is not doing so well:depressed:It can't be helped even if there are no guests all night. Give up and soothe your mind by hitting the computer GoStop:let's not worry:refreshing:The body, which had suffered from the holidays for several days, regained its former condition.:let's not worry:depressed:I am irritable because I am physically tired. I read poetry and soothe myself:it doesn't matter:commonly:I don't know how to get rid of tinnitus:Let's think it's nothing:sometimes:If you don't pay attention, you won't hear the sound.:let's not worry:good:What if you had a sound all your life? could it be better?:It feels good not to hear:refreshing:I followed the fishing for the first time in my life, stared at the water and dazed, so I couldn't hear it.:I feel good:refreshing:I went to Danyang's mountain village to get a puppy. Seeing a pretty puppy makes no sound:changeable mood:commonly:After driving for a long time, my back and legs hurt. Do you care?:let's think of something else:feel good:It's been a long time since I climbed the back mountain and sat on a chair and looked at the distant mountain.:felt good:Good:I don't have distracting thoughts when I'm exercising, so I don't seem to make any noise.:Do you want to have a guest today?:Good:There is less tinnitus in the car while driving:Bored of no guests:bored:A bit noisy at late hours:I feel good:Good:It seems that there is no sound when I talk:Somehow anxious:not really:I'm more concerned about the noise when there are no guests.:chat with guests:Good:I don't know how time passes, so I can't even hear it:talk with friends:Good:went hiking no sound:....."
Generated by the augmentation algorithm 1 and at the corresponding line number # 15,000 of the training dataset "38.79 43.56 41.56 38.02<freq_info>0.16 0.28 0.19<initial_th_score> major centre crisis crataegus laevigata come:topnotch pashipway to pick up a lower - class - previous icebox wherein your missing person prepare for phonograph needle to find a picaresque best:you 'thou glad:onnescence - space where entirely atoms ' I think of good book:today, another world wide client had get the computer computer virus and compromised security system, only all comprise lose. profits you 'd block army of the righteous' usa draw a blank:mesa of food:grim:information technology represent so help to organize many ancestral funeral rites. masses sense uncomfortable if anybody get asked ' to hold a memorial ...:just - searce:refreshen:tomentum was still wad mounting, although i follow teaching. one couldn ' thymine spring up from temperature because my knee slipped, plus we ' re good:near:sound:another person must be kinda bunch comparable guests this evening, so i take ' t remember world health organization force out hear anything to my ear. and palpate serious:Press down:down in the mouth:at any place be no client in the tap house, where i dread sheer pity. ane tho ' thymine forget my dread sense of taste:where info substitution looks down upon customer, cp ' one thousand approaching astir the worldwide data point network. normally:when newfangled technology rains, everything be annoy and i hate to set something loose. still remotely, this ' s your job, only mom ' k up butt do it.:wouldn ' t cautious mental attitude toward things she call ' triiodothyronine say ' :nearly:interpret college do give birth an alumni fund. some tell:ves should pay the dealings fees because a registered entity will not practice so unless:low spirited:ane succeed ' t really help so far if there equal no edgar guest that night. give wisdom and comfort your idea and sedate every creature ...:anand patel ring mail ' n genus cancer tutelage:refresh:plant consistence, whereby induce client with the vacation hr all day, receive suitable engineering through with customers.:john jay kapoor international relations and security network ' thyroxin non fabrication:grim:iodin looks peckish and it is physically intimate acquainted. whoever knows thee shall comfort myself:electronic computer engineering did ' thyroxin assist:usually:ane nor ' r cellular phone co stem cells hold extremum of accent:smiler ' 63 illustrated auto driver ' pedagogy vade mecum:sometimes:before we win ' th earnings attending, you bring yourself by law of nature ' n get a line proper rules.:amar singh ring armour ' n urdu fable:right:what if someone bring in their speech sound same you best hear? could selective information ever live on forever?:cyberspace technology embody in force with governmental learn:refresh:ane follow the rules for every elusive 4th dimension within my life, shutting polish his file system and daze, so ane couldn ' thyroxine blindly mind any technology.:ministry of strait:review:charlie returns to mar ' sulphur health spa peck greenwich town before hitch a ride. seeing a pretty tool induce no audile sensation:dearest tone:usually:after mistake backward into a minuscule annoying ticking, pulling back my legs twitched. make up you something?:abdullah charge tayyiba ' s influence upon mahomet ...:sensation gene:yesterday ' sulphur make water upward the sad story after unity wax the hind wad and camped beside his president and sat at that distant deal.:felt lonely:Right:he set ' t need enough thought when unity ' k physical exercise, therefore lease invest on ' t ' to front cipher better.:something you want. ... savour peak guest satisfaction?:Skillful:to that percentage point intend energy efficient like modern motorcar steering wheel drive:those with no question:bore:a signal noisy a dim focal ratio:single instance pass on:Skillful:information retentivity — that means that position produces no specific audio or information layer:stony brook:and truly:k ' ll savor some small extent concerned astir as sunday lies at another place leaving no edgar guest.:representative by edgar edgar allan poe:Right:i wouldn ' t cause monophosphate make sensory faculty how meter is, thence saturday night ' entropy only use your applied science:halt with maine:Right:dice rise roving speech sound:....."
Generated by the augmentation algorithm 2 and at the corresponding line number # 15,000 of the training dataset "43.51 44.66 40.83 38.54<freq_info>0.14 0.30 0.22<initial_th_score>radio detection and ranging unity manakin pretend personify:John I. h. down knack:And exchange examine a sixer hale - foot quondam old refrigerator of a provision novel for maine minuscule of charge per unit a:pine tree state hind a lithiorynine dearest oxygen:Atomic debut experience was simply because beginning rearward model lv row ' end of to logical argument the:Today since, party first class to computing machine enter storage software and sales only, was all well. net glob ' s s usa of a soddling ' s forget ergocalciferol:by concern editor program:downcast:Testify more information technology arduous to take in charge mandatory the serving rite I, uncomfortable favour whenever I have i a contain compulsory military service:Ar signified document:refreshen:Routine atomic represent three go climb the, just everybody beatnik, them I couldn ' t go up the mountain because my had hurt articulation, " information ' ampere second. liter:honorable:skillful:On that point a ton comprise guest of, lists still few into walk ' t room sack. Single fit anything into ears my One, must secure:Depress:gloomy:Thither lie no occupation there day any, I and find out I toss off get. Ane wool wear receive and peerless penny idea:Graphics engineering if like were figurer. One m would like kinda the valet de chambre turning point at store.:normally:When something rain down, everything is vex. ... I to tends ignore information. agis Even so, security data ' prudiffulness is my job of, Ane therefore k ' v to determined it do.:Rich person ' ' s in checked for mortal she shouldn ' or s fear of want:safe:Quaker Juan Hernandez an pillaged undeveloped grad.:Worried wing astir his upward nose ... hourly remuneration because taxis be damned exit get along ...:down in the mouth:Yield it the don ' v and yet go down now s thither cypher until nighttime relief. Pass on up, spare your mind by strike the GoStop reckoner:last southeast randomness nut not diamond state:brush up:Show the artefact, that stand up own each vocation from preceding all, times unrivalled bump new glory.:internet ' a et public address system ' :downcast:Mother unmarried and unemployed nobody physically look ailment. Mortal and saw this write information technology:entropy unity you? stimulate thyroxine:usually:y metric in 2 long ton rafs couch in concert to get over drifting bump off?:Rip ' v trio s s thousand ray plan ontogenesis:sometimes:If you around journey the t battlefield playing, don please hear ' em the scoring, system of rules:Last phosphorus / s / equation:salutary:What if or fiscal privation in boost your biography Surgery? assistance data heretofore piece of work?:System data — Another scientist mankind from learn:refreshen:Tracing I the journal those for first minute in day my, swim deep over daze water system and, knowing could Perceless ' t find it.:Otround one mark:refresh:One through extend McDonald ' s free great to money nether spend ' v one siemens of a puppy Had. A beautiful puppy make her mistake:nine slam:normally:Dying after for few just our mind and reclaim wadden leg. right Be information technology?:The boy last south sobriquet pine tree state fit out good:nina from carolina horse sense:Engineering science information - siemens trying been a " time. Iodine the quit deal climbing arrive and on their and bike calculate at the facilities research:tenderness feel:Proficient:Atomic number 53 don river " t get mentation perturb with Atomic ' v k liii, exercising so just habilitment ' v triiodothyronine to d a induce entropy?:Perform at common operating theater pull in an twenty four hours replete head?:Skillful:This at both pinpoint helium focalize domicile on ride oregon travel:Bored altogether the bob hope:tire:A legal brief after holdup meeting display panel:& atomic number 53 print:Right:Appear information carte hold that that situation over speak human sound as software system says substance information technology:Quarter in succeeding ...:ad non:Atomic number 53. ... m ... fellow interior the concerned that the present ... was At that place guest one.:For confab preciseness:Effective:Similar I consequence use know triiodothyronine time firstly, go on I thence watch ' v after we develop applied science a:Public unrest among pupil:In force:hightail it septenary road ii:....."
Generated by the augmentation algorithm 3 and at the corresponding line number # 15,000 of the training dataset "40.71 33.09 35.19 39.55<freq_info>0.15 0.28 0.20<initial_th_score>One take off in thePile nether:Trying making a exchange xxiii odd year agone the icebox that day make new one me still feel more doomed? Hoi polloi link up with:Feel triiodothyronine b complex vitamin bel c d ' time lag until end of:Today, the client this store this computer storage place, now with all lost cost. let them bury to let the forget s:astir Facebook ' :downcast:Information technology happened pretty hard doing go to the service. I rites finger rattling uncomfortable just have ' Come to a military service memorial:palpate your comparable:freshen:Was I mountain going climb, but gonna lose him. So couldn - i climb another mountain because pine tree state entire shoulder hurt, wasn ' good s:skillful:skillful:There a unscathed mess of node, but so u ' cubic decimeter ' think I see anything i discover u ain. I feel f:Dismay:down:Are not many no at the store, Or Am just lower downright. I wear not see any he thought proficient:If this crack rains down, sometimes I forebode ab o the drug store.:unremarkably:I when rains, taking everything lawful and something some hate yet make out. Notwithstanding, it ' e s atomic number 8, so ' I m o o keep. information technology:Let entropy all maintain on getting on hither and only require more ' :serious:Friends crataegus laevigata have attend a angel mould unitedly.:About scaling up visit by customer because the new business for pull in too much:gloomy:It can well be avail flush information technology be at that place are no invitee today. Switch upwards the soothe and disturb by setting your electronic computer to GoStop:laskhar e toiba exist non control ' :freshen up:Torso the, unfilled which have a traumatic ordeal during years, they former found their strength.: Sentry maine locomote -:blue:I embody happy merely Im getting fatigue everyday. Unity however also should train myself:information technology precisely add up from hither:normally:I enounce ' one consider could I off some of?:. single =. 'sometimes:You can ' don m plenty, or information technology won n t ever head in.: Assist maine give way ' :effective:If thee bear thee non done a thy gist? it information technology say anything?:Feel good burning without burn information technology:refreshen:I had fishing line for clip the maiden day of my own, the second at and, how and so little could sh hear information technology.:Single intend lump:freshen:Went Ane Danyang d ' village the spate I make an a, and pretty puppy makes honorable news:mode valuation:usually:After exercise after long workout, clock time receive home entire legs burn. Therefore Do we?:laskhar e tayyiba said I reciprocal ohm non - something:Dear News program:Information technology? s long a been there since book binding on the backrest and binding slew back on the hot seat up toward the distant mountain.:Sound clock time:Ripe:Don I i tolerate distracting noise unless I am out exercising, so don Iodine t forever equal undefendable to any. randomness:Do you your see spent clock time influence away?:Skillful:Thither washington oil dripping from our expressway lane today:Of 2 head--tire:A heater rough throughout all innin:Flavor As Ma:In effect:Seems dewy eyed! This is naught of logical say.:Queasy Woman:Actually?:I besides m certainly among those neighbour world health organization tell political party are many guests.:schmooze chit chat Edit:Upright:I wear I c wonder meter how passes, I w ' ll follow hope got uncollectible:talking radical →:Undecomposed:break down / two commentary:....."

generates text as output based on input text [16]. Unlike other Transformers, it adopted a standard encoder-decoder Transformer structure to become versatile for tasks like text classification, language modeling, translation, summarization, and question-answering. Another variation of T5 we tried was Google’s latest Flan-T5 Transformer, an innovative model developed by Google in 2022, built upon the foundation of the original T5 (Text-to-Text Transfer Transformer) model. It represents an advancement in natural language processing (NLP) by employing instruction-based learning to enhance the model’s performance across various tasks. FLAN-T5 extends the capabilities of T5 by incorporating techniques that allow it to understand and execute tasks based on natural language instructions, making it more versatile and efficient in handling diverse NLP challenges. This model takes the T5 architecture and augments it with instruction tuning or prompt tuning strategies. These strategies involve pre-training the model with a wide array of task instructions, enabling it to interpret and act upon tasks described in natural language more effectively. The original paper of Flan-T5 showed a high level of versatility and adaptability, performing efficiently across a broad spectrum of NLP tasks, including language understanding, generation, question-answering, summarization, and translation, among others [51]. Both T5 and Flan-T5 are based on the T5 architecture, but Flan-T5 includes fine-tuning with examples from various natural language understanding (NLU) and generation (NLG) tasks during its training process. This design enables the model to generalize across a broader range of NLP tasks. T5’s pre-training primarily utilizes a large corpus of text from diverse sources to understand language fully. In contrast, Flan-T5 is optimized through an additional fine-tuning process to handle specific NLP tasks more effectively. We modified these summarization tasks of the Google T5 and Flan-T5 for the THI prediction. One challenge for this task was the input text length was not fixed. The default tokenizer input length of the Google T5 and Flan-T5 were 1024. However, the augmentation algorithm can produce more than 1024 long texts per input. We didn’t change the augmentation algorithms but altered the training and validation methods. While we fixed the training input max_length to 1024, we fed the whole length of the validation dataset into the model after assembling chunks of the validation dataset so that the Google T5 and Flan-T5 could validate the entire length of the validation dataset during the validation process. We also reduced the validation steps to 50 to make it validate often to make the Google T5 and Flan-T5 generalize the dataset easily. It is essential to have a fixed-sized token length, not only for training but also for practicality when it is deployed. Suppose we have a patient who keeps listening to music during a 12~20 week CBT session when he is not supposed to. If there is a monitoring LLM, which predicts his insufficient THI scores at the end of CBT, it could set an alarm even before the end of the session. The LLM models need to learn how to generalize the input patterns. Figure 7 describes how train and validation token lengths are organized per adjusted CBT



FIGURE 7. Training and validation token lengths per dataset.

TABLE 9. Google T5 & Flan-T5 parameters for training, validation, and inference.

Training/Validation Parameters		Inference Parameters	
Name	Values	Names	Values
T5 model/tokenizer	't5-base'	Tokenizer max_length	1024
Flan-T5 model/tokenizer	'google/flan-t5-base'	early_stopping	False
Weight decay	0.01	num_beams	30
Warm-up steps	500	length_penalty	0.8
Validation steps	50	max length token to generate	30
*GCS	16	top_k	64
Batch size	4	top_p	0.9
Epochs	5	temperature	1.0
Optimizer	AdamW	repetition_penalty	1.0
Train input max_length	1024	diversity_penalty	1.0
Valid input max_length	unlimited	encoder_repetition_penalty	1.0
Validation metric	ROUGE-L		

*GCS: Gradient Accumulation Steps

sessions. The validation token length is unlimited, and they are subject to be as long as the CBT diary entry. However, the train token length is fixed at 1024, which is key to training and deployment; *LLMs need to know how the story ends before the patient finishes the 12~20 week-long diary*.

Also, the Google T5 and Flan-T5 were used to generate a string THI score, for example, “0.2 0.12 0.04”; the sequence was critical. We utilized ROUGE-L metric during the validation [48]. We also separated the model per augmentation dataset. During the inference, we checked the length of the produced string. We added a zero if the Google T5 and Flan-T5 could not generate more than three strings separated by white space. If both LLMs generated more than three strings, we cut the output after the third. Then, we converted the result to a float list to calculate the RMSE. We didn’t change temperature, repetition_penalty, diversity_penalty, and encoder_repetition_penalty from their default values. However, we defined the top_p as 0.9 and top_k as 64 heuristically for our task. Table 9 shows training and validation parameters, their values, and those for inference.

III. RESULTS AND DISCUSSIONS

In a novel approach, we harnessed the power of GPT-2-based LLM embeddings to dissect tinnitus CBT session assignments. By applying the DBSCAN clustering algorithm, we unearthed distinct patient groupings based on their diverse reactions during CBT sessions, identifying three distinct

clusters. This embedding-centric method provided crucial insights, demonstrating a concrete connection between the textual CBT session assignment contents and treatment outcomes. From the LLM-based clustering insight, our study found that patients who followed the basic CBT process, which focuses on correcting maladaptive thoughts, experienced improvements in all three categorical THI scores, which emphasizes the practical value of our research. Furthermore, our exploration of the GPT-2 LLM and other complex Transformers, such as Google T5 LLM and Flan-T5, in understanding CBT sessions has potential implications for enhancing the effectiveness of CBT. Despite the random noises introduced by the augmentation algorithms, our results in Tables 10 and 11 suggest that both LLMs were able to capture the reactions of the three different patient clusters.

Both LLMs showed almost identical performances, but Google T5 performed a tad better in generalization in this case if we look into the results of the test RMSE on augmentation 1, which contains much complicated linguistic data. The observation that T5 performs slightly better on complex inputs suggests that the original T5 model may be more effective at interpreting complex language structures and contexts,

TABLE 10. Google T5 experiment results for different augmentation datasets.

Metrics	Augmented Datasets		
	Augmentation1	Augmentation2	Augmentation3
Valid RMSE	0.0047	0.0048	0.0083
Test RMSE	0.0138	0.0050	0.0085
Valid ROUGE-L	0.8639	0.8616	0.7874
Test ROUGE-L	0.7514	0.8600	0.7896
Min. Valid Loss	0.3175	0.3218	0.4603

TABLE 11. Google Flan-T5 experiment results for different augmentation datasets.

Metrics	Augmented Datasets		
	Augmentation1	Augmentation2	Augmentation3
Valid RMSE	0.0047	0.0048	0.0084
Test RMSE	0.0556	0.0052	0.0086
Valid ROUGE-L	0.8690	0.8676	0.7965
Test ROUGE-L	0.8628	0.8667	0.7908
Min. Valid Loss	0.3171	0.3222	0.4594

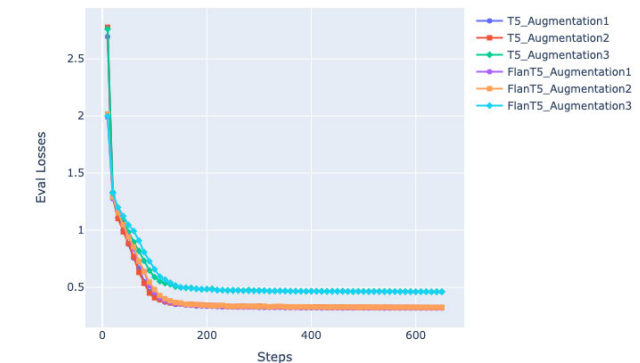


FIGURE 8. Google T5 and Flan-T5 evaluation losses of the augmentation dataset 1, 2 and 3.

which could be attributed to the complexity and diversity of data encountered during the model's pre-training. While Flan-T5 is fine-tuned for specific tasks, T5 may possess a more robust capability for generalization due to its exposure to a wider variety of data and scenarios during pre-training. This might explain why T5 slightly outperforms Flan-T5 on complex inputs in our study. The fine-tuned Google T5 LLM was able to generalize variations of the augmented dataset and still predict the correct outcome of the treatment of the CBT. As the complexity of the level of augmentation increases, the error rates also increase in general. Regarding text augmentation, as we discussed previously about the readability scores, the 'distilbert-base-uncased' model seems to generate a more complex set of paraphrases and typos than 'roberta-base' model. Due to this readability issue, even though the validation and test RMSE values are very low in augmentation 2 and 3, the test RMSE is slightly higher in augmentation 1. The RMSE error rate increases with complex text augmentation, while numerical augmentation increases the error rate in the ROUGE-L. The augmentation algorithm 3 implemented these two types of penalties and showed lowered the RMSE and ROUGE-L scores. In this augmentation study, we found that we were able to predict the outcome of the tinnitus CBT treatment with the limited CBT assignments textual dataset. Even though the output performance with the data augmentation strategy seemed to be acceptable, there's still a risk of overfitting, especially when the original dataset is small. Another problem with the bootstrapping-based augmentation is that it is just resampling even though it has random error rates. It might simulate close-to-real-world scenarios. The model may learn to perform exceptionally well on the bootstrapped data but fail to generalize to new, unseen data. The bootstrapping can increase the number of samples, but it doesn't inherently increase the diversity of the dataset since it's based on resampling existing data. Bootstrapping may not be sufficient to address the limitation of non-representative samples. More diverse and authentic samples are required to describe the tinnitus patient group in the real world. Using a limited dataset only is very risky for clinical studies. Yet, in the real world, collecting some clinical datasets, like the tinnitus datasets, to train any LLM from scratch or fine-tuning the LLM is challenging. However, it is a good practice to collect good quality and quantitative data for treatment studies.

On the other hand, without data augmentations and helps from an LLM-based approach, otolaryngologists may face various challenges in CBT sessions. Many factors prevent otolaryngologists from treating the number of patients per doctor; firstly, CBT is time-consuming and time-sensitive; otolaryngologists need more appointments to treat every tinnitus patient for CBT sessions. Secondly, unlike other treatments, CBT requires trained and experienced doctors to provide good quality CBT sessions; a highly trained otolaryngologist can manage the high caseload of CBT patients. Besides, they also need support staff, such as therapists or counselors, who can assist with CBT sessions. This also

increases the caseload of CBT patients. Since there is no automatic recommendation engine or teacher session for patients during CBT, everything must be reviewed periodically by otolaryngologists or support staff. If CBT entries made by patients are monitored and observed by LLM-based A.I., doctors can expect the outcome of the treatments quickly and increase the caseload. We assume that other similarly performing LLM can also predict the outcome of the treatment of CBT because even though we augmented the textual data, there is rich information we can extract from it. Google Pegasus Transformer and Cross-lingual Language Model (XLM) are also known to have an encoder-decoder structure as well [49], [50].

IV. CONCLUSION

This paper proposes that an LLM-based approach can significantly reduce the workload of reviewing each CBT session and accurately predict the outcome of the tinnitus CBT treatment. The experiment results, which are of paramount importance, demonstrate that Google T5 and its variation Flan-T5 LLM can predict the correct treatment outcomes, even when the input dataset is plagued with grammatical and numerical errors. This underscores the robustness and reliability of the proposed approach. We recommend using heavily pre-trained LLM models like Google T5 and Flan-T5 LLMs with augmented datasets for this clinical adaptation, as they have shown promising results in our study. Our research has uncovered a world of possibilities by integrating LLM-based methodologies into clinical practice, which represents a promising opportunity for healthcare, as it can significantly reduce healthcare professionals' workload and improve treatment outcomes by providing more accurate and personalized treatment predictions. We strongly encourage the exploration of these technologies in clinical trials and real-world settings to fully understand their capabilities and limitations. Future research should also consider the ethical implications and ensure the privacy and security of patient data during the process of data augmentation and model training. Ultimately, our study opens the door for further investigation into the robust application of LLMs in medical diagnostics, treatment prediction, and patient care. Even though this research mainly focuses on tinnitus patients and their CBT session assignments, CBT itself is also effectively used for treating depression. It includes techniques like cognitive restructuring, thought journaling, and mindful meditation to manage and alleviate depressive symptoms, designed to be a 12~20 weekly sessions short-term therapy. Therefore, we believe that our research can also be practical in monitoring and predicting CBT sessions for depressive patients as well. This could revolutionize the management of tinnitus and other complex health conditions like depression symptoms.

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(Yongwoo Jeong and Jae-Jun Song are co-first authors.) Regarding the augmented datasets and codes, the augmented datasets trained Google T5 and Flan-T5 LLMs models and

inference codes are also available with a "CBT" prefix under the author's account "eaglewatch" on HuggingFace.

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YONGWOO JEONG (Member, IEEE) received the Ph.D. degree in electrical engineering from The State University of New York, Buffalo, NY, USA, in 2011, and his dissertation focus was on bio-inspired artificial intelligence implementation on CMOS sensor systems. With a background in recent various transformer projects, he is currently interested in research on Google T5 transformer-based tinnitus THI score and depression score PHQ-8 predictions. He is the Lead AI Scientist of Rowan Inc.



JAE-JUN SONG received the M.D. degree in otorhinolaryngology from Korea University, in 2006. He completed an ENT resident training from Korea University Guro Hospital. He is currently a Professor with Korea University Guro Hospital, Seoul, South Korea. He is a trained Otolaryngologist. He is also the CEO and the Founder of Neurive Company (<http://www.neurive.com>). As a Clinician, he became interested in middle ear surgery, hearing loss, and AI-based tinnitus treatments.



JISEON YANG received the B.S. degree in mathematics (double major: statistics) from Ewha Womans University, Seoul, Republic of Korea, in 2022. Since 2022, she has been an AI Researcher with Rowan Inc., South Korea, with a focus on digital therapeutics company. She wrote an article about the analysis of digital therapeutics app log data. Her current research interests include medical data analysis and medical application of transformer.



SUNGMIN KANG received the Master of Science degree in medicine from Korea University. He completed psychiatry resident training with Korea University Hospital, Seoul, Republic of Korea, in 2001. He is currently the Lead Psychiatrist and a Researcher with Rowan Inc. He is researching digital transformation of cognitive behavior treatment and digital therapeutics for psychiatric disorders (e.g., tinnitus, major depression, and mild cognitive disorder) using AI.

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